UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which structure chooses between two alternative paths?
   A. array
   B. if...else
   C. struct
   D. for

2. How many times does a while loop execute when its condition is false initially?
   A. Two
   B. Until the user stops it
   C. Zero
   D. One

3. What does the % operator return?
   A. The quotient
   B. The absolute value
   C. The square root
   D. The remainder of integer division

4. Which phase of the program development life cycle defines what the program must do?
   A. Analysis
   B. Coding
   C. Testing
   D. Maintenance

5. What happens when an array index is outside its valid range?
   A. The compiler always rejects the program
   B. The array becomes a string
   C. Undefined behaviour may occur
   D. C++ automatically resizes the array

6. Which input pattern stops when the user enters -1?
   A. A sentinel-controlled loop
   B. A function prototype
   C. A fixed for loop
   D. A switch statement

7. Which term describes a finite sequence of precise steps that solves a problem?
   A. Variable
   B. Algorithm
   C. Compiler
   D. Array

8. Which data type stores a whole number?
   A. string
   B. char
   C. int
   D. float

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9. Which loop is guaranteed to execute its body at least once?
   A. for
   B. while
   C. if
   D. do...while

10. Which declaration creates a floating-point variable named price?
   A. float price;
   B. int price;
   C. bool price;
   D. char price;

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade: 80-100 A, 60-79 B, 50-59 C, 40-49 D, below 40 F.

a) Draw a flowchart for the grade classification.

(5 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(8 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following nested loop:

```
for (int i = 1; i <= 3; i++) {
   for (int j = 1; j <= i; j++) {
      cout << j;
   }
   cout << endl;
}
```

a) Trace the program and show the exact output.

(7 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

(8 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

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PART C (30 MARKS)

QUESTION 1

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

QUESTION 2

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

END OF QUESTION PAPER